Informatica® Metadata Command Center
April 2026

Oracle Business Intelligence Sources

Informatica Metadata Command Center Oracle Business Intelligence Sources
April 2026

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Preface

Read *Oracle Business Intelligence Sources* to learn how to register and configure Oracle Business Intelligence sources in Metadata Command Center as catalog sources. After you configure a catalog source, you extract metadata and then view the results in Data Governance and Catalog.

CHAPTER 1

Introduction to Oracle Business Intelligence catalog sources

You can use Metadata Command Center to extract metadata from a source system.

A source system is any system that contains data or metadata. For example, Oracle Business Intelligence Enterprise Edition Presentation Server is a source system from which you can extract metadata through an Oracle Business Intelligence catalog source with Metadata Command Center. A catalog source is an object that represents and contains metadata from the source system.

Before you extract metadata from a source system, you first create and register a catalog source that represents the source system. Then you configure capabilities for the catalog source. A capability is a task that Metadata Command Center can perform, such as metadata extraction, lineage discovery, data profiling, data classification, or glossary association.

When Metadata Command Center extracts metadata, Data Governance and Catalog displays the extracted metadata and its attributes as technical assets. You can then perform tasks such as analyzing the assets, viewing lineage, and creating links between those assets and their business context.

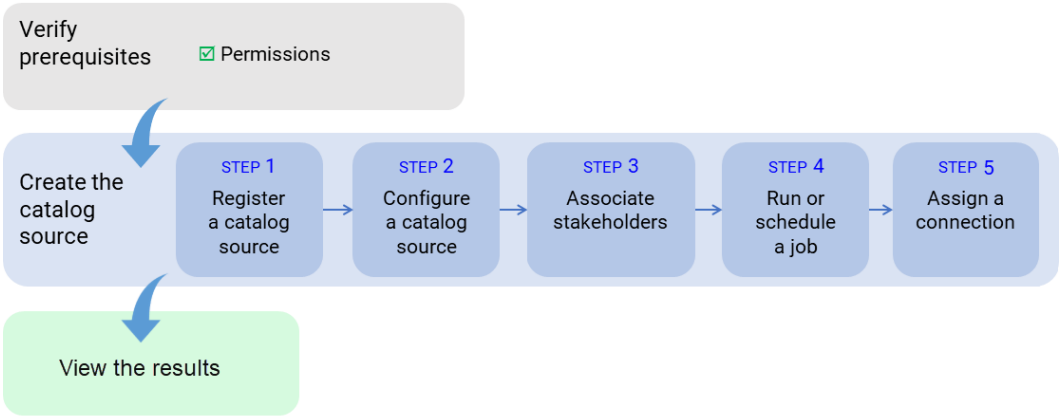
The following table describes the capabilities of the catalog source:

Capability	Description
Data Classification	Data classification is the process of identifying and organizing data into relevant categories based on the functional meaning of the data. Classifying data can help your organization manage risks, compliance, and data security.
Glossary Association	You can associate terms that are in the glossary with technical assets to provide user-friendly business names to technical assets. Glossary Association automatically associates glossary terms with technical assets or recommends glossary terms that you can manually associate with technical assets in Data Governance and Catalog.

Extraction and view process

To extract metadata from a source system, configure the catalog source and run the extraction job in Metadata Command Center. Then view the results in Data Governance and Catalog.

The following image shows the process to extract metadata from a source system:



After you verify prerequisites, perform the following tasks to extract metadata from Oracle Business Intelligence:

1. Register a catalog source. Create a catalog source object, select Oracle Business Intelligence, and specify values for connection properties.
2. Configure the catalog source. Specify the runtime environment and configure parameters for metadata extraction. Optionally, add filters to include or exclude source system assets from metadata extraction. You can also configure other capabilities such as data profiling and quality, data classification, or glossary association.
3. Optionally, associate stakeholders. Associate users with technical assets, giving the users permission to perform actions determined by their roles.
4. Run or schedule the catalog source job.
5. Optionally, if the catalog source job generates referenced asset objects, you can assign a connection to referenced source system assets.
   You can view the lineage with object references without performing connection assignment. After connection assignment, you can view the objects.

After you run the catalog source job, you view the results in Data Governance and Catalog.

About the Oracle Business Intelligence catalog source

You can use the Oracle Business Intelligence catalog source to extract metadata from the Oracle Business Intelligence Enterprise Edition Presentation Server.

Oracle Business Intelligence is a business intelligence tool to gather, store, and visualize enterprise data. You can create and configure an Oracle Business Intelligence catalog source to extract metadata from the Oracle

Business Intelligence Enterprise Edition Presentation Server. Metadata Command Center extracts report objects that include presentations, business, and physical models.

Extracted metadata

You can extract specific metadata from source systems that you connect to with an Oracle Business Intelligence catalog source.

Metadata Command Center extracts the following metadata from an Oracle Business Intelligence source system:

- Alias
- Alias Column
- Alias Key Column
- Alias Table
- Answers Report
- Business Model
- Business Model and Mapping
- Catalog
- Column
- Condition
- Criteria
- Criteria Column
- Criteria Hierarchical Column
- Dashboard
- Dashboard Page
- Dashboard Prompt
- Database
- Data Set
- Data Source
- Derived Logical Column
- Derived Presentation Column
- Element
- External Schema
- Folder
- Group
- Hierarchy
- Key
- KPI
- Link
- List of Values
- Logical Column
- Logical Dimension

- Logical Fact Table
- Logical Key Column
- Logical Level
- Logical Table
- Measure
- Parameter
- Parameter Answer
- Parameter Value
- Physical Column
- Physical Key Column
- Physical Table
- Presentation Column
- Presentation Fact Table
- Presentation Hierarchy
- Presentation Level
- Presentation Table
- Presentation Variable
- Prompt
- Prompt Answer
- Publisher Data Model
- Publisher Report
- Report
- Report Field
- Repository
- Repository Variable
- Session Initialization Block
- Session Variable
- Static Variable
- Subject Area
- Value
- Variable

CHAPTER 2

Before you begin

Before you create a catalog source, ensure that you have the information required to connect to the source system.

Perform the following tasks:

- Assign the required permissions.
- Download the Oracle Business Intelligence RPD repository file.
- Convert the Oracle Business Intelligence RPD repository file to an XML or UDML file.
- Install the Informatica Secure Agent on a Windows or Linux machine.
- Import SSL certificate to the JRE folder of the Informatica Secure Agent.
- Copy the following files to a directory on the Informatica Secure Agent machine:
  - Oracle Business Intelligence RPD variable values file that contains a list of RPD variable values in the key=value format.
  - Converted XML or UDML metadata files.
  - Data sources XML file that defines the Oracle Business Intelligence Publisher data source connections.
- If your database is SSL-enabled, specify appropriate values for the Trust Store file and Trust Store password fields in the configuration properties.

  Note: Test Connection to an SSL-enabled database fails if either the trust store file or trust store password value is not specified in the configuration properties.
- Create endpoint catalog sources for connection assignment.

Verify permissions

To extract metadata and to configure other capabilities that a catalog source might include, you need account access and permissions on the source system. The permissions required might vary depending on the capability.

Permissions to extract metadata

To extract Oracle Business Intelligence metadata, you need account access and read permissions to the Oracle Business Intelligence Enterprise Edition Presentation Server.

Permissions to perform data classification

You can perform data classification with the permissions required on the source system to perform metadata extraction.

Permissions to perform glossary association

You can perform glossary association with the permissions required on the source system to perform metadata extraction.

Get Oracle Business Intelligence source information

Get the values of the connection properties that you need to configure from the Oracle Business Intelligence Enterprise Edition Presentation Server administrator.

Note: You don't need to create a connection object for Oracle Business Intelligence. You provide this information when you configure the catalog source.

The following table describes the properties that you need:

Property	Description
Version	Select the version of the Oracle Business Intelligence Enterprise Edition Presentation Server from where you want to import metadata. Select one of the following options to specify the version: - Auto Detect. Select this option to let Metadata Command Center detect the version of Oracle Business Intelligence. - OBIEE 10.x. Select this option if you use the Oracle Business Intelligence 10.x version. - OBIEE 11.x. Select this option if you use the Oracle Business Intelligence 11.x or 12.x versions.
Server URL	URL to connect to the Oracle Business Intelligence Enterprise Edition Presentation Server. Format is: http://<host name>:<port number>/analytics/saw.dll If you use SSL, ensure that Metadata Command Center trusts the server certificate of the Oracle Business Intelligence Presentation Server. **Note:** If single sign-on is enabled for the user, replace analytics with analytics-ws in the server URL. Format is: http://<host name>:<port number>/analytics-ws/saw.dll
Login User	User name to access the Oracle Business Intelligence Enterprise Edition Presentation Server. Ensure that the user name you use has the necessary permissions to import metadata.
Login Password	Password associated with the user name.
Metadata File	Absolute path to the XML or UDML metadata file located on the Secure Agent machine. **Note:** - For Oracle Business Intelligence version 10.x, you must convert the RPD file to a Universal Database Markup Language (UDML) file. - For Oracle Business Intelligence versions 11.x and 12.x, you must convert the RPD file to XML format. Metadata Command Center doesn't support the UDML file format for Oracle Business Intelligence 11.x and 12.x versions. For information on how to convert RPD files to XML and UDML files, see "Convert the Oracle Business Intelligence RPD repository file to an XML or UDML file" on page 12.

Download the Oracle Business Intelligence RPD repository file

The Oracle Business Intelligence Administration tool natively stores metadata in an RPD repository file. To convert the RPD file to an XML or Universal Database Markup Language (UDML) file, you need to download the RPD file.

1. Start a command prompt in the console of the machine where Oracle Business Intelligence Presentation Server is installed.
2. Navigate to the `bitools\bin` folder of your WebLogic server domain.
   For example, you can run the following command for Windows operating systems:

   ```
   cd Oracle_Home\user_projects\domains\bi\bitools\bin
   ```

3. Run the **downloadrpd** download repository command.
   To run the **downloadrpd** command, you need the data model launder script. Use `datamodel.cmd` for Windows and `datamodel.sh` for Linux operating systems.

   For example, you can run the following command to download the live RPD repository for Windows operating systems:

   ```
   datamodel.cmd downloadrpd -O repository.rpd -SI ssi -U weblogic -P weblogicpsw
   ```

   You are prompted to choose a password for the local RPD file.

Convert the Oracle Business Intelligence RPD repository file to an XML or UDML file

After you download the RPD repository file, you can convert the file to XML or UDML format. The steps to convert the RPD file to an XML or UDML file differ based on the Oracle Business Intelligence version.

Oracle Business Intelligence version 12.2.1.4

Run the **biserverxmlgen** command to generate the XML file.

Use `biserverxmlgen.cmd` for Windows and `biserverxmlgen.sh` for Linux operating systems.

For example, you can run the following command to generate the XML file for Windows operating systems:

```
biserverxmlgen.cmd -R repository.rpd -P Admin123 -O repository.xml -N -Q -8
```

Note: Metadata Command Center doesn't support the UDML file format for Oracle Business Intelligence version 12.x.

For more information, see the corresponding Oracle documentation.

Oracle Business Intelligence version 11.x

1. Run the initialization script to configure a command prompt or shell window that is initialized to your Oracle instance.
   For recent 11.x versions, it is `bi-init.cmd` for Windows and `bi-init.sh` for UNIX operating systems.

   For example, you can run the following initialization script for Windows operating systems:

   ```
   ORACLE_INSTANCE\bifoundation\OracleBIApplication\coreapplication\setup\bi-init
   ```

   For older 11.x versions, it is `nqinit.bat` for Windows and `nqinit.sh` for UNIX operating systems.

For example, you can run the following initialization script for UNIX operating systems:

```
/u01/wls/oracle/j2ee/bi/bifoundation/server/bin/nqinit.sh
```

You might have to configure the following environment variables for UNIX operating systems:

- export ORACLE_HOME=/u01/wls/oracle/j2eeexport
- export DOMAIN_HOME=/u01/wls/oracle/j2ee/user_projects/domains/mod

2. Run the **biserverxmlgen** command to generate the XML file.
   Use `biserverxmlgen.cmd` for Windows and `biserverxmlgen.sh` for Linux operating systems.

   For example, you can run the following command to generate the XML file for Windows operating systems:

   ```
   biserverxmlgen -R paint.rpd -P Admin123 -O paint.xml -N -Q -8
   ```

Note: Metadata Command Center doesn't support the UDML file format for Oracle Business Intelligence version 11.x.

For more information, see the corresponding Oracle documentation.

Oracle Business Intelligence version 10.x

Run the **nQUDMLGen** command in the `$OracleBIHome$\server\bin` directory to convert the RPD file to a UDML file.

For example, you can run the following command to generate the UDML file for Windows operating systems:

```
QUDMLGen.exe -U Administrator -P Administrator -R paint.rpd -O paint.udml -N -Q -8s
```

For more information, see the corresponding Oracle documentation.

Import the SSL certificate to the Informatica Secure Agent machine

If SSL is enabled on the Oracle Business Intelligence source system, import the SSL certificate to the JRE folder of the Informatica Secure Agent installation directory.

Complete the following steps to import the SSL certificate:

1. Download the SSL certificate from the Oracle Business Intelligence installation.
2. Copy the SSL certificate file to any location on the Informatica Secure Agent machine.
3. Identify the Java version for the Secure Agent.
   You can identify the Java version from the following files:

   - `<Informatica Secure Agent installation directory>\apps\agentcore\agentcore.log`
     Search for "AgentCore JRE version".
   - On Windows operating systems, open the lcm-env.bat file from one of the following locations:
     - `<Informatica Secure Agent installation directory>\apps\agentcore\<latest version>\.lcm`
     - `<Informatica Secure Agent installation directory>\apps\DIS\<latest version>\.lcm`

- On Linux operating systems, open the lcm-env.sh file from one of the following locations:
  - `<Informatica Secure Agent installation directory>/apps/agentcore/<latest version>/.lcm`
  - `<Informatica Secure Agent installation directory>/apps/DIS/<latest version>/.lcm`

4. Open a command prompt from the following directory:
   `<Informatica Secure Agent installation directory>\apps\jdk\<latest version>\jre\bin`
5. Run the following command to import the SSL certificate:

   ```
   keytool -import -alias <alias name> -keystore <path to cacert file> -file <absolute path to SSL certificate>
   ```

   Note: The Java certificate file is named cacerts and is located in the following Java directory: `\jre\lib\security\cacerts`

   For example, you can run the following command on Windows operating systems:

   ```
   keytool -import -alias aliasname -keystore "C:\data\devprod\jdk\jre\lib\security\cacerts" -file "C:\data\devprod\filename.crt"
   ```

6. Restart the Secure Agent.

Create endpoint catalog sources for connection assignment

You can perform connection assignment to reference source systems to view source system objects. Before you perform connection assignment, create catalog sources for each reference source system that you want to connect to.

An endpoint catalog source represents a source system that the catalog source references. Before you perform connection assignment, create catalog sources and run the catalog source jobs for source systems that your catalog source references.

CHAPTER 3

Create catalog sources in Metadata Command Center

Use Metadata Command Center to configure a catalog source for Oracle Business Intelligence and extract metadata.

When you configure a catalog source, you define the source system that you want to extract metadata from. Configure filters to include source system metadata before you run the job. Optionally, configure other capabilities, such as lineage discovery, data profiling and quality, data classification, relationship discovery, and glossary association.

To provide stakeholders access to technical assets, you can assign access through stakeholder roles. You can also associate technical assets extracted from the catalog source to asset groups. If your catalog source references other source systems, you can create a connection assignment to the endpoint catalog source to view complete lineage.

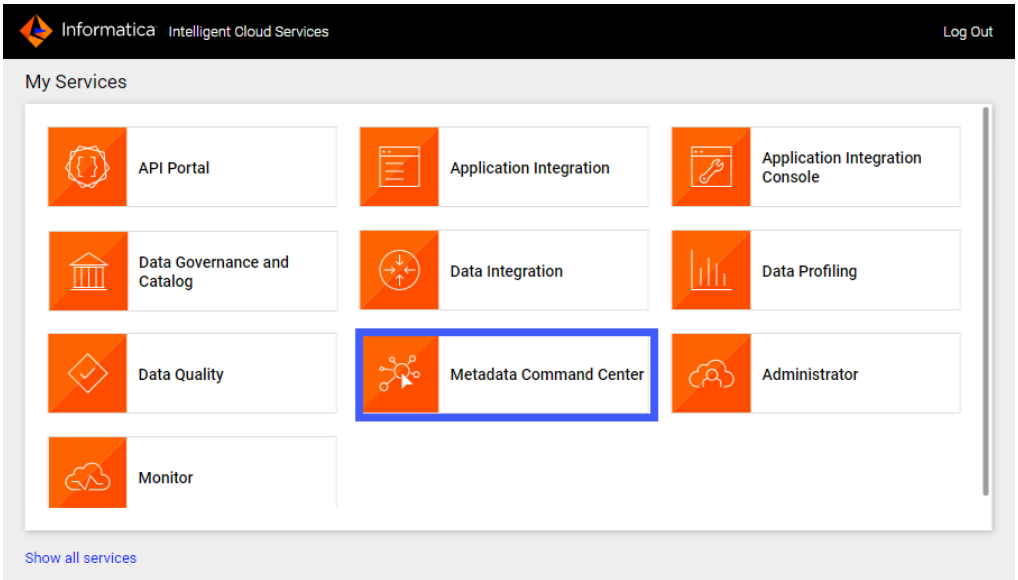
Step 1. Register a catalog source

When you register a catalog source, provide general information and connection values.

1. Log in to Intelligent Data Management Cloud.

   The **My Services** page appears.
2. Click **Metadata Command Center**.

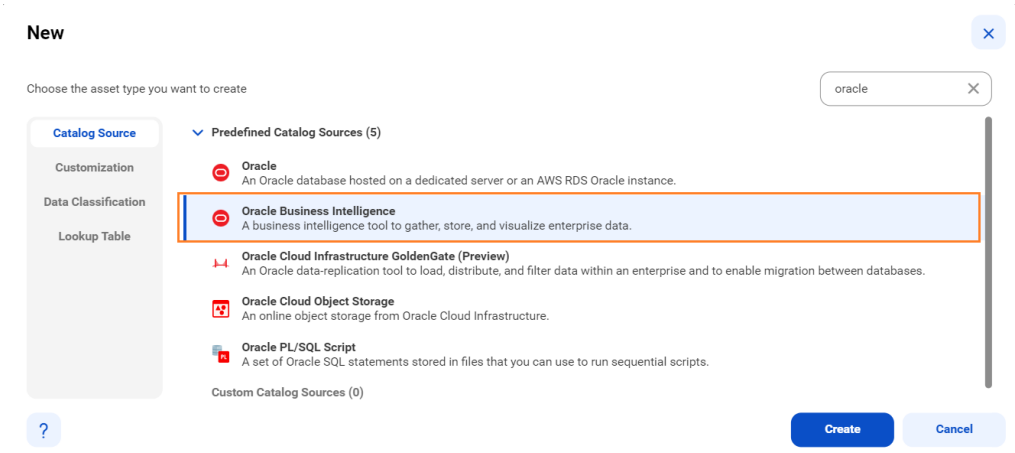
   The **Metadata Command Center** box appears on the **My Services** page.



The Metadata Command Center home page appears.

3. Click **New**.
4. Select **Catalog Source** from the list of asset types.
5. Select Oracle Business Intelligence from the list of catalog source types.

   The following image shows the Oracle Business Intelligence catalog source:



6. Click **Create**.

   The **New Catalog Source** page opens.
7. In the **General Information** section, enter a name and an optional description for the catalog source.

   Note: You can rename a catalog source after you create it, but to apply the change to all associated objects you must rerun the metadata extraction job.

   After you save the catalog source, you can update the description in Metadata Command Center and Data Governance and Catalog. The update appears only in the service in which you update it.
8. In the **Connection Information** area, enter the Oracle Business Intelligence connection information.

The following table describes the properties to configure:

Property	Description
Version	Select the version of the Oracle Business Intelligence Enterprise Edition Presentation Server from where you want to import metadata. Select one of the following options to specify the version: - Auto Detect. Select this option to let Metadata Command Center detect the version of Oracle Business Intelligence. - OBIEE 10.x. Select this option if you use the Oracle Business Intelligence 10.x version. - OBIEE 11.x. Select this option if you use the Oracle Business Intelligence 11.x or 12.x versions.
Server URL	URL to connect to the Oracle Business Intelligence Enterprise Edition Presentation Server. Format is: http://<host name>:<port number>/analytics/saw.dll If you use SSL, ensure that Metadata Command Center trusts the server certificate of the Oracle Business Intelligence Presentation Server. **Note:** If single sign-on is enabled for the user, replace analytics with analytics-ws in the server URL. Format is: http://<host name>:<port number>/analytics-ws/saw.dll
Login User	User name to access the Oracle Business Intelligence Enterprise Edition Presentation Server. Ensure that the user name you use has the necessary permissions to import metadata.
Login Password	Password associated with the user name.
Metadata File	Absolute path to the XML or UDML metadata file located on the Secure Agent machine. **Note:** - For Oracle Business Intelligence version 10.x, you must convert the RPD file to a Universal Database Markup Language (UDML) file. - For Oracle Business Intelligence versions 11.x and 12.x, you must convert the RPD file to XML format. Metadata Command Center doesn't support the UDML file format for Oracle Business Intelligence 11.x and 12.x versions. For information on how to convert RPD files to XML and UDML files, see "Convert the Oracle Business Intelligence RPD repository file to an XML or UDML file" on page 12.

9. Click **Next**.

The **Configuration** page appears.

Step 2. Configure capabilities

When you configure the Oracle Business Intelligence catalog source, you define the settings for the metadata extraction capability.

The metadata extraction capability extracts source metadata from external source systems. You can also configure other capabilities that the catalog source includes.

You can save the catalog source configuration at any point after you enter the connection information. After you save the catalog source, you can choose to run the catalog source job. To run the job once, click **Run**. To run metadata extraction and other capabilities on a recurring schedule, configure schedules on the **Schedule** tab.

Configure metadata extraction

When you configure the Oracle Business Intelligence catalog source, you choose a runtime environment and enter configuration parameters for metadata extraction.

1. In the **Connection and Runtime** area, choose a serverless runtime environment or the Secure Agent group where you want to run catalog source jobs.

   Note: Serverless runtime environment options are available if the catalog source works with a serverless runtime environment.

2. Choose to retain, delete, or deprecate objects that are deleted from the source system in the catalog with the **Metadata Change Option**.

   - **Retain**. Retains objects that are deleted from the source system in the catalog. If you update or add a filter, the catalog retains objects extracted from the previous job and extracts additional objects that match the current filter. Objects deleted from the source system are not deleted from the catalog. Enrichments added on deleted objects and relationships are retained.
   - **Delete**. Deletes metadata from the catalog based on objects deleted from the source system and changes you make to the filter. Enrichments added on deleted objects and relationships are also permanently lost. Objects renamed in the source system are removed and recreated in the catalog.
   - **Deprecate**. When objects are deleted from the source system, the catalog changes the lifecycle of the imported objects to Obsolete. The lifecycle of objects also changes to Obsolete when you add exclusion filters or reduce the scope so that objects extracted in earlier jobs are no longer included. For example, the first job includes the sales schema. The second job also includes the sales schema, but excludes the sales.transactions table. In this case, the second job marks the sales.transactions object as Obsolete, even though the first job includes it. This does not impact enrichments applied to deprecated objects and relationships. Objects renamed in the source system are removed and recreated in the catalog. When you run the catalog source job again for other capabilities such as data classification, relationship discovery, or glossary association, the job doesn't consider obsolete objects. Obsolete objects remain in the catalog until they are purged when you run a **Purge Obsolete Objects** job on the **Explore** page.

   Note: You can also change the configured metadata change option when you run a catalog source.

3. In the **Filters** area, define one or more filter conditions to apply for metadata extraction:

   a. Select **Yes** to view filter options.

   b. From the Include Metadata list, select **Include Metadata** to include metadata based on the filter parameters.

   c. From the object type list, select **Objects**.

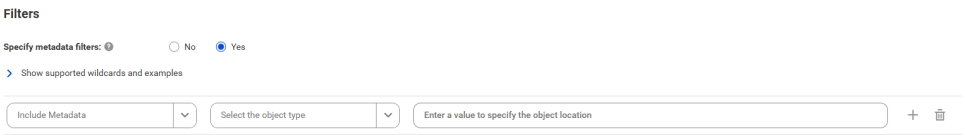
   d. Enter the filter values.

      Filters can contain the following wildcards:

      - Question mark. Represents a single character.
      - Asterisk. Represents multiple characters or empty text.

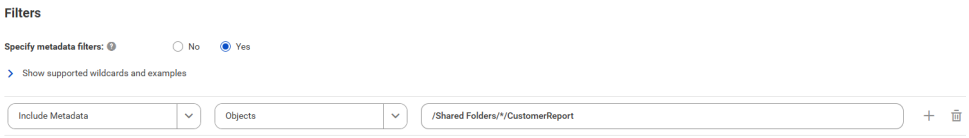
      For path hierarchies, use the '/' separator. Filters are case-sensitive. To extract metadata from My Folders, the path in the filter value must start with: /User Folders/My Folders/

      The following image shows the filter condition options:



e. To define an additional filter with an OR condition, click the **Add** icon.

The following image shows that the filter includes metadata from all objects named CustomerReport located across folders in the specified Oracle Business Intelligence repository:



4. In the **Configuration Parameters** area, enter configuration parameters.

The following table describes the properties that you can enter:

Property	Description
Variable Values File	Absolute path to the Oracle Business Intelligence Repository RPD variable values file located on the Secure Agent machine. The file contains a list of RPD variable values in the following format: key=value
Data Sources File	Absolute path to the data sources XML file located on the Secure Agent machine. The file defines the Oracle Business Intelligence Publisher data source connections.
Optimize for Large Models	Choose whether you want to optimize metadata import for large files. Select one of the following options: - True. Optimizes the import of metadata for large Oracle Business Intelligence repository models. - False. Imports metadata of the entire repository model. This might result in a high consumption of memory. If you select True, Metadata Command Center doesn't import metadata for the following assets: - Foreign keys - Joins - Relationships - Logical foreign keys In addition, Metadata Command Center doesn't store expression tree objects with lineage links.
Incremental Import	To configure how you want to extract metadata, select one of the following options: - True. Extracts only the metadata that changed since the last metadata extraction job. - False. Extracts the complete metadata.
Worker Threads	Number of worker threads to process metadata asynchronously. You can leave the value empty or enter a positive integer. If you don't enter a value, Metadata Command Center computes and assigns a value between 1 and 6 based on the JVM architecture and the number of available CPU cores on the Secure Agent machine. Default values are between 1 and 6. If you enter an invalid value, Metadata Command Center shows a warning and uses the value 1. If your machine has more memory, you can specify a higher value to process more metadata asynchronously. **Note:** Specifying a higher value might impact performance of the system.
Trust Store File	Path to the Oracle Business Intelligence SSL trust store file. Enter the location of the directory on the Secure Agent machine where you copied the trust store file. For example, `<Informatica Secure Agent installation directory>:\<FolderName>\<truststorefilename.jks>`

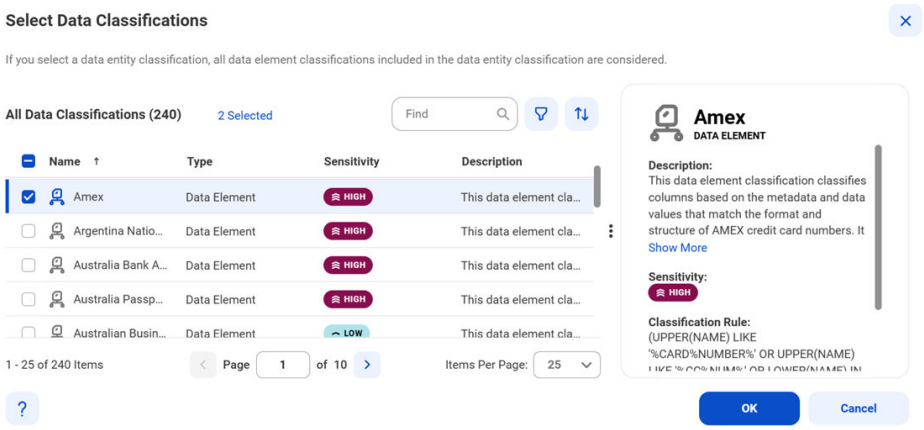
Property	Description
Trust Store Password	Password to access the contents of the trust store file.
Miscellaneous Options	Enter values for additional extraction runtime options in the following format: -key <value> Start each value with a hyphen followed by parameters. Separate multiple values with a space.

5. Configure additional capabilities for the catalog source by clicking on the tabs.

Configure data classification

Enable the data classification capability to identify and organize data into relevant categories based on the functional meaning of the data.

1. Click the **Data Classification** tab.
2. Select **Enable Data Classification**.
3. Choose one or both of the following options:
   - **Generated Data Classifications**. CLAIRE automatically generates data classifications for the data elements.
   - **Data Classification Rules**. Choose from predefined or custom data classifications.
     1. Click **Add Data Classification**. The **Select Data Classifications** dialog box appears.



     2. Select the data classifications that you want to use.
     3. Click **OK**.

Configure glossary association

Enable the glossary association capability to associate glossary terms with technical assets, or to get recommendations for glossary terms that you can manually associate with technical assets in Data Governance and Catalog.

Metadata Command Center considers all published business terms in the glossary while making recommendations to associate your technical assets.

1. Click the **Glossary Association** tab.

2. Select **Enable Glossary Association**.
3. Select **Enable auto-acceptance** to automatically accept glossary association recommendations.
4. Specify the **Confidence Score Threshold for Auto-Acceptance** to set a threshold limit based on which the glossary association capability automatically accepts the recommended glossary terms.

   Note: Specify a percentage from 80 to 100. If the score is higher than the specified limit, the glossary association capability automatically assigns a matching glossary term to the data element.
5. Select **Enable Below-threshold Recommendations** to receive glossary association recommendations below the auto-acceptance threshold. If you enable auto-acceptance, you can enable below-threshold recommendations to receive glossary recommendations below the auto-acceptance threshold.
6. Specify the **Confidence Score Threshold for Recommendations** to set a threshold based on which the glossary association capability makes recommendations

   If you enable auto-acceptance, specify a percentage from 80 to the selected auto-acceptance threshold. You can accept or reject the recommended glossary terms that fall within this range in Data Governance and Catalog.

   If you disable auto-acceptance, specify a percentage from 80 to 100 inclusive.
7. Choose to automatically assign business names and descriptions to technical assets. You can then choose to retain existing assignments and only assign business names and descriptions to assets that don't have assignments, or allow overwrite of existing assignments.

   By default, existing assignments are retained.
8. Optional. Choose to ignore specific parts of data elements when making recommendations. Select **Yes** and enter prefix and suffix keyword values as needed.

   Click **Select** to enter a keyword. You can enter multiple unique prefix and suffix keywords. Keyword values are case insensitive.
9. Optional. Choose specific top-level business glossary assets to associate with technical assets. Selecting a top-level asset selects its child assets as well. Select **Top-level Glossary Assets** and specify the assets on the **Select Assets** page.
10. Optional. Choose to use abbreviations and synonym definitions from lookup tables for accurate glossary association. Select **Yes** to enable, and then click **Select** to upload a lookup table.
11. Click **Next**.

    The **Associations** page appears.

Step 3. Associate stakeholders and asset groups

Associate users or user groups within a stakeholder role as stakeholders for technical assets in Data Governance and Catalog. Also, you can choose to assign technical assets extracted from the catalog source to asset groups. You can then use access policies to control permissions on assets that are assigned to asset groups.

Verify that the administrator assigned users and user groups to the stakeholder role that you want to associate with technical assets.

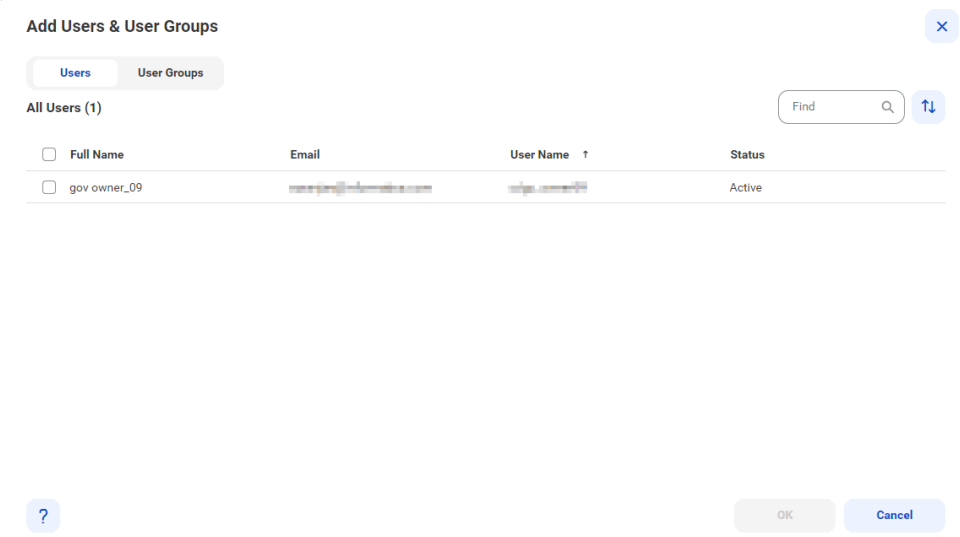
1. To associate users or user groups as stakeholders with technical assets extracted from the catalog source, perform the following steps:
   a. On the **Associations** page, click **Stakeholders**.

b. Select **Assign Stakeholders**.

c. Select a stakeholder role.

d. Click **Select** to add users and user groups from the stakeholder role as stakeholders for the technical assets.

The **Add Users & User Groups** dialog box displays a list of users and user groups assigned to the selected stakeholder role.



e. Select one or more users or user groups to assign as stakeholders for the technical assets, and click **OK**.

Only the selected users and user groups belonging to the specified stakeholder role are granted the permissions to technical assets.

f. To assign users or user groups from another stakeholder role, click **Add** and then repeat the steps.

2. To assign asset groups to technical assets extracted from the catalog source, perform the following steps:

a. On the **Associations** page, click **Asset Groups**.

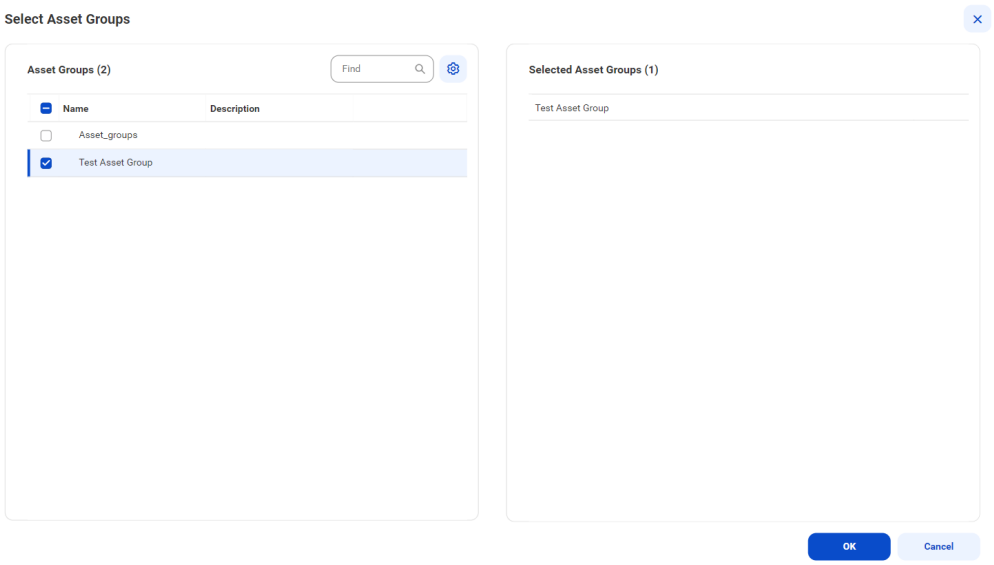
b. Select **Assign Asset Groups**.

c. Click **Select**.

The **Select Asset Groups** dialog box displays the list of asset groups.

If you enabled an access policy that includes an asset group, you can only view assets that belong to that asset group.

3. Select the asset groups to which you want to assign technical assets extracted from the catalog source, and click **OK**.



4. Choose to save and run the job or to schedule a recurring job.
   - To save and run the job, click **Save** and then **Run**.
   - To schedule a recurring job, click **Next** to open the **Schedule** page.

Step 4. Run or schedule the job

Choose to run a catalog source job manually, or configure it to run on schedule.

Note: You can't run multiple jobs simultaneously.

You can choose to perform a full or an incremental metadata extraction. A full metadata extraction extracts all objects from the source to the catalog. An incremental metadata extraction extracts only the changed and new objects since the last successful catalog source job run. Incremental metadata extraction doesn't remove deleted objects from the catalog and doesn't extract metadata of code-based objects if applicable.

When you run an incremental metadata extraction job with a filter to include metadata from objects, the job extracts only the objects that have the latest timestamp since the last successful job.

Note: If incremental metadata extraction is available for the catalog source, the option is always visible, but incremental metadata extraction jobs can run only after a scheduled full metadata extraction job. Manual jobs do not enable the incremental metadata extraction option.

Run the job manually

Click **Save** to save the catalog source and click **Run**. On the **Run Catalog Source Job** window, click **Run** to run the job.

You can override the capabilities that you selected while configuring your catalog source on the **Configuration** page. The first time you run the catalog source job, the metadata extraction capability is mandatory. From the second run onwards, you can choose to override the configured metadata change option. You can retain, delete, or deprecate objects that are deleted from the source in the catalog. For subsequent runs of the catalog source job, the metadata extraction capability is optional.

Note: You can choose incremental metadata extraction for subsequent runs only after one full metadata extraction job completes successfully. Incremental metadata extraction jobs run with the **Retain** metadata change option even if you set the option to **Delete** or **Deprecate** in the catalog source.

Note: To run a catalog source job, you need permissions on the connection to the source system. To run a catalog source job for catalog sources that reference other source systems, you need permissions on the connections for all the reference source systems.

Run the job on a schedule

You can choose to run metadata extraction and other capabilities on a recurring schedule. You can't choose incremental metadata extraction and full metadata extraction in the same schedule. To create a schedule for incremental metadata extraction, you must have completed at least one full metadata extraction job successfully. If not, first create a schedule for a full metadata extraction.

If an incremental metadata extraction is scheduled to run when the last run details aren't available, the job first performs a full metadata extraction, followed by incremental metadata extraction on subsequent runs.

For example, this can happen in the following scenarios:

- You create schedules for both incremental metadata extraction and full metadata extraction, but schedule the incremental extraction to run before the first full metadata extraction job.
- You create schedules for both incremental metadata extraction and full metadata extraction, but delete the full metadata extraction schedule before its first run.

1. On the **Schedule** tab, select **Run on Schedule**.
   The **Schedule** configuration page opens.
2. Click the checkbox corresponding to each capability that you want to include in the schedule.
3. Enter the start date, time zone, and the interval at which you want to run the job.
4. You can manage additional schedules using the following options:
   - To create a new schedule, click the **Add** button.
   - To delete a schedule, click the **Delete** button.
   - To enable or disable a schedule, click the **Enable Schedule** toggle button.

   Note: You can create a maximum of one schedule per capability that you enable. If you purged a catalog source or did not run the metadata extraction job, the catalog source job runs metadata extraction before running other scheduled capabilities.

   Note: To create a schedule, you need permissions on the connection to the source system. If you lose permissions on the connection after you create a schedule, the scheduled jobs continue to run.
5. Click **Save** to save the schedule.

Monitor job status

After the job runs, you can monitor the status of the job on the **Overview** page of the job.

For more information about job monitoring, see *Administration*.

Step 5. Assign reference catalog source connections to endpoint catalog source objects

When you run the catalog source job, if the catalog source references another source system, a reference catalog source and connection get created that point to the reference source system. To view the complete

lineage for your catalog source, you can perform connection assignment from the reference catalog source connection to the objects in the reference source system. A referenced source system might be a database such as Oracle. You must first create and run an endpoint catalog source that connects to the reference source system.

Before you assign a connection, ensure that you have created and run an endpoint catalog source for each reference source system.
Note: If the source schema contains case-sensitive tables or if the reference objects contain multiple objects with the same name in different cases, perform case-sensitive connection assignment to get correct lineage.

If you enabled the lineage discovery capability for your catalog source, you can either curate the CLAIRE recommended endpoint objects on the **Related Catalog Sources** tab or assign connections manually.

For more information about related catalog sources and lineage discovery, see *Lineage discovery* in the *Administration* help.

1. On the **Configure** page, select the **Lineage** tab, and then select the **Lineage Discovery** tab. On the **Catalog Sources** panel, select the required catalog source and click the **Assign Connections** tab.

   The **Assign Connections** tab displays a list of assigned and unassigned connections along with details for each connection. Use filters to view the connections based on the connection names. Click the **Add Filter** menu to add filters.

2. Select the connection to the reference source system and click **Assign**.

   The connection name appears prefixed to the reference catalog source name on the **Hierarchy** tab of your catalog source in Data Governance and Catalog.

   The **Assign Connection** dialog box appears with a list of recommended objects from the endpoint catalog sources. Click **All** to view all endpoint catalog source objects.

3. In the **Assign Connection** dialog box, select one or more endpoint objects to assign to the selected connection and click **Assign**.

   You can filter the list in the **Assign Connection** dialog box by name, type, or endpoint.

   You can assign Oracle relational databases as reference source systems. To create a connection assignment, the endpoint object can belong to the Schema or Database class type.

When you click **Assign**, Metadata Command Center creates links between matching objects in the connected catalog sources, and it calculates the percentage of matched and unmatched objects. The higher the percentage of matched objects, the more accurate the lineage that you view in Data Governance and Catalog.

CHAPTER 4

View results in Data Governance and Catalog

After Metadata Command Center runs a job, you can view the results in Data Governance and Catalog where the catalog source and its elements are called technical assets. You can view a catalog source as a hierarchy. Expand each technical asset to see its components.

When referenced source systems are connected to a catalog source, you can expand the hierarchy to see details about the technical asset's component elements.

You can view the data lineage of an asset contained within a catalog source to see individual elements such as data sources, calculations, and filters. When you view data lineage, you can see the individual upstream elements that contribute data or expressions to each component of a data flow or catalog source.

View metadata extraction results

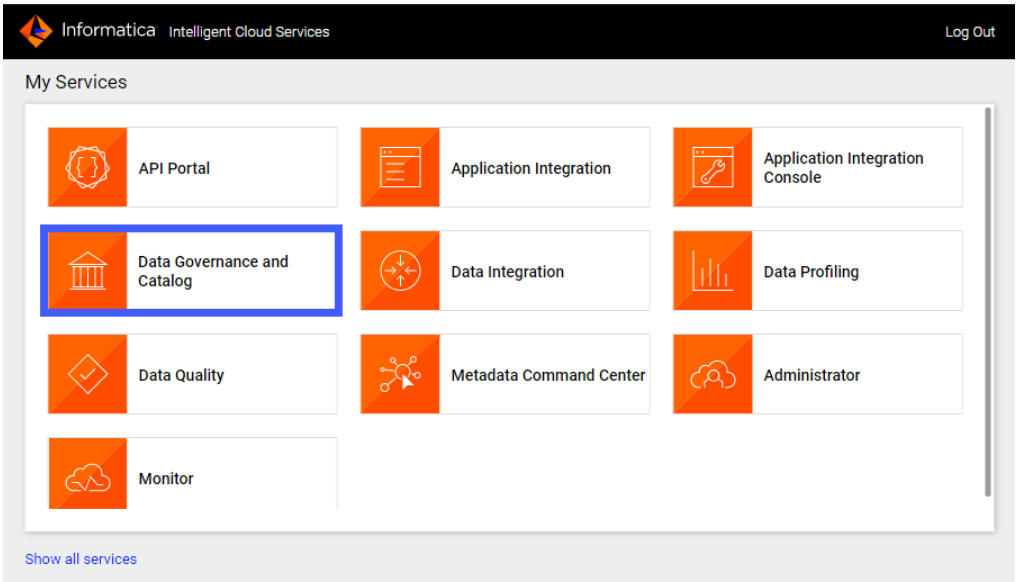
After a job runs in Metadata Command Center, view the results in Data Governance and Catalog. You can view details about source system contents in a hierarchical structure and trace data lineage.

1. Log in to Intelligent Data Management Cloud.

   The **My Services** page appears.

2. Click Data Governance and Catalog.

   The Data Governance and Catalog box appears on the **My Services** page.



3. On the Data Governance and Catalog home page, click the number in the **Technical Assets** panel.

   The **Technical Assets** page opens.

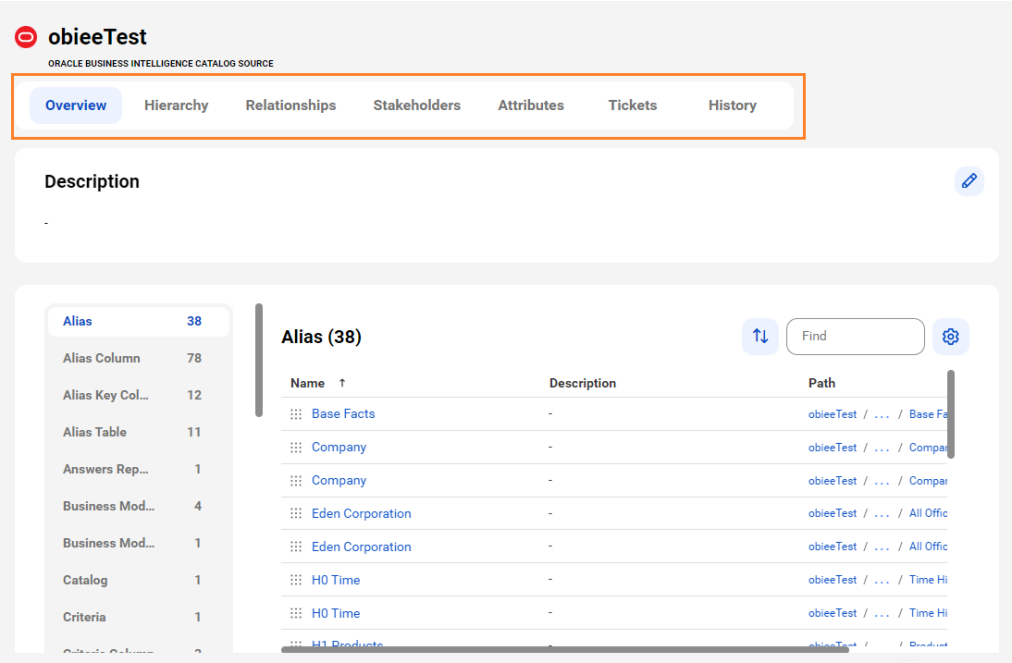
4. Select **Catalog Source** in the **Filter** list.

   The list of catalog sources opens.

5. Search for the catalog source from which you extracted metadata, and click the name.

   The **Overview** tab of the asset opens.

   The following image shows a sample asset page:



6. View the asset from different perspectives by clicking on the tabs.

For more information about working with assets, see Working with Assets in *Data Governance and Catalog* help.

View data lineage

Data lineage is a visual representation of the flow of data across the systems in your organization. Lineage depicts how the data flows from the system of its origin to the system of its destination.

Data lineage views are available for technical assets in the catalog source. You can view lineage at the catalog source, data set, or data element level.

The lineage at the catalog source level shows how data flows from one catalog source to another. The lineage at the data set and the data element levels show how other technical assets such as files or tables contribute to the selected asset.

If linking catalog sources is available for your catalog source, you can use Metadata Command Center to generate data lineage based on rules or by generating automated lineage with CLAIRE. You can choose source and target catalog sources and objects to link and generate lineage.

To determine whether linking catalog sources is available for your catalog source, navigate to the **Configuration** tab of the **Link Catalog Sources** page. The catalog source must appear in the list of source and target catalog sources.

For information about linking catalog sources, see *Link catalog sources* in the Administration help.

View lineage at the catalog source level

The catalog source level shows how data flows from one catalog source to another with the lineage aggregating data from the data set and data element levels.

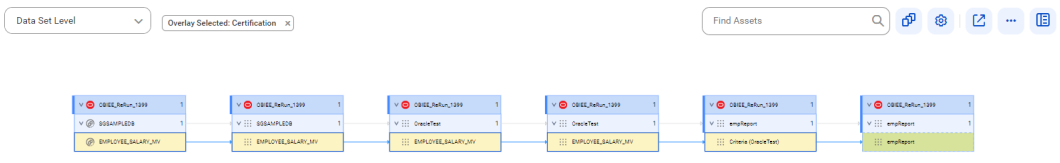
To view data lineage at the catalog source level, open a technical asset, click the **Lineage** tab, and then verify that the level is set to **Catalog Source Level**.

View lineage at the data set level

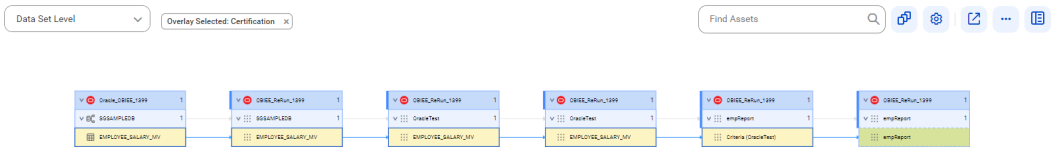
The data set level displays individual sets of data in the data flow.

To view lineage at the data set level, open a technical asset, click the **Lineage** tab, and then verify that the level is set to **Data Set Level**.

The following image shows data set level lineage where the empReport report of an Oracle Business Intelligence source system gets data from the EMPLOYEE_SALARY_MV reference table of a reference source system before connection assignment:



The following image shows data set level lineage where the empReport report of an Oracle Business Intelligence source system gets data from the EMPLOYEE_SALARY_MV table of an Oracle source system after connection assignment to the Oracle source system:

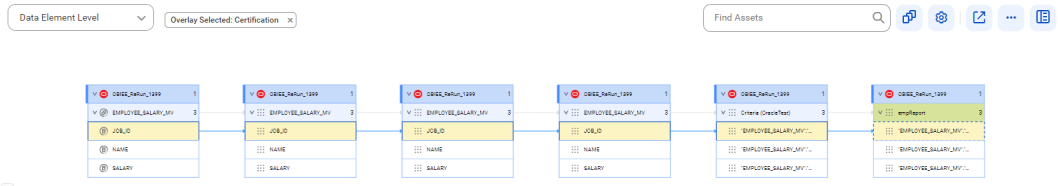


View lineage at the data element level

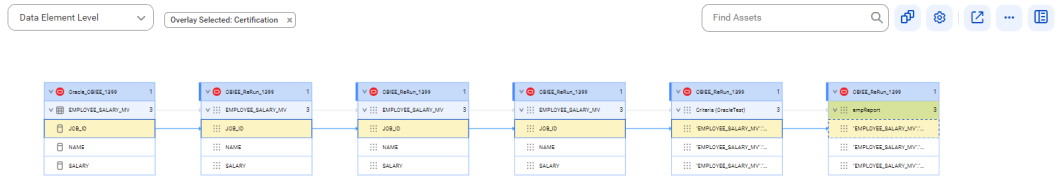
The data element level displays details of the data set level. At the data element level, you can see the input sources for expressions or commands and calculations or transformations on the data.

To view data lineage at the data element level, open a technical asset, click the **Lineage** tab, and then verify that the level is set to **Data Element Level**.

The following image shows data element level lineage where the "EMPLOYEE_SALARY_MV"."JOB_ID" report field of the empReport report of an Oracle Business Intelligence source system gets data from the JOB_ID reference column of the EMPLOYEE_SALARY_MV reference table before connection assignment:



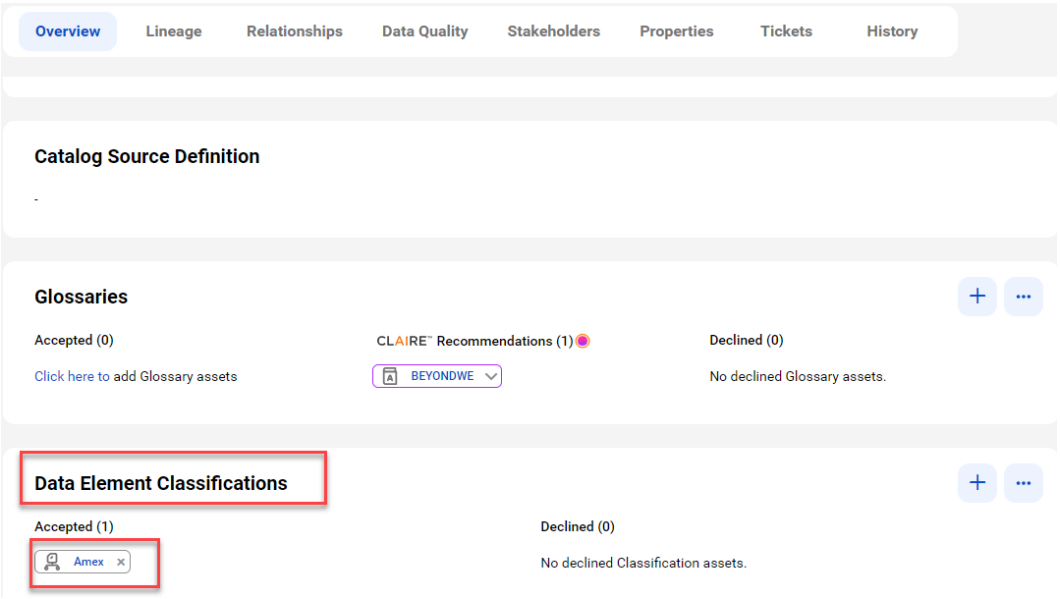
The following image shows data element level lineage where the "EMPLOYEE_SALARY_MV"."JOB_ID" report field of the empReport report of an Oracle Business Intelligence source system gets data from the JOB_ID column of the EMPLOYEE_SALARY_MV table of an Oracle source system after connection assignment to the Oracle source system:



View classified data

When you add data classification rules to a catalog source in Metadata Command Center, the system identifies the columns and tables that match the rules and displays one or more matched data classifications on the column or table asset pages in Data Governance and Catalog.

The following image shows a column asset page with the inferred data element classifications that match the column data and metadata:



For more information about data classification assets, see *Asset Details* in the Data Governance and Catalog help.

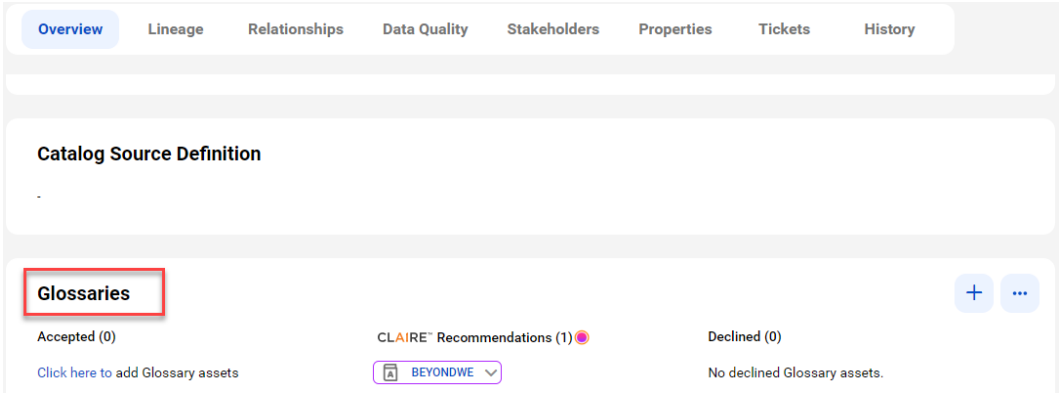
View glossary associations

When you enable the glossary association capability for a catalog source in Metadata Command Center, you can view the accepted glossary assets in Data Governance and Catalog.

The **Overview** tab for a technical asset in the catalog source displays glossary assets in the Accepted and CLAIRE Recommendations sections.

The **Glossaries** panel shows the automatically accepted and CLAIRE® recommended terms.

The following image shows a sample asset page:



APPENDIX A

Managing catalog sources

You can update, copy, delete, purge, and run a catalog source. You can also grant configuration permissions for catalog sources to different roles in your organization.

On the **Catalog Sources** page of the **Explore** page, you can select a catalog source to see its overview and related catalog sources. The **Overview** tab displays the capabilities, filters, stakeholders, asset groups, and schedules enabled for a catalog source. The **Related Catalog Sources** tab displays a list of related catalog sources that are either assigned manually using connection assignment or are curated based on CLAIRE recommendations using the lineage discovery capability. You can view related catalog sources and accept or reject CLAIRE recommendations for catalog sources that are run with the lineage discovery capability.

Cloning a catalog source configuration

You can clone the configuration of a catalog source to create another catalog source with the same configuration.

Note: Cloning doesn't copy the content of the catalog source.

1. In Metadata Command Center, go to the **Explore** page.
2. Select **Catalog Sources** from the menu.
3. You can clone a catalog source in one of the following ways:
   - Hover your mouse over the catalog source that you want to clone and click **Clone** from the **Action** menu.
   - Open the catalog source that you want to clone and click **Clone**.

   Note: By default, the cloned catalog source has the same name suffixed with _Clone. You can change the name if needed.
4. If needed, modify the configuration of the catalog source by updating the parameters on each tab.

   Cloning doesn't copy the content of sensitive fields. If the catalog source contains sensitive fields such as Client Secret, enter the values.
5. Click **Save**.

Deleting or purging a catalog source

Delete or purge a catalog source based on your business requirements.

When you delete a catalog source, you delete all the configuration specific to that catalog source, along with all the metadata that was extracted and any enrichment made to the associated metadata. Purging a catalog source deletes all the metadata that was extracted, along with any enrichment made to the associated metadata.

Note: Before you delete or purge a catalog source, manually unassign the connections that are assigned to that catalog source.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source that you want to delete or purge.
4. Click the **Action** menu, and select **Delete** or **Purge**.

   Note: You can't delete or purge a catalog source if the catalog source job is in the running state.
5. To confirm delete or purge, click **OK**.

   The catalog source is triggered to delete or purge. You can monitor the status of the job on the **Monitor** page.

   Note: If the deletion job succeeds, the catalog source disappears from Metadata Command Center. If it fails, some associated objects might still be present in Data Governance and Catalog. Such catalog sources continue to appear in the list of catalog sources in Metadata Command Center, but are marked as **Deleted**. Delete the catalog sources to remove them along with any remaining objects present in Data Governance and Catalog.

Purging obsolete objects from a catalog source

You can purge obsolete objects from a catalog source.

When you configure or run a catalog source, you can choose to deprecate objects that are deleted from a source system. If you delete objects from the source system or make changes to the filter, the objects imported into the catalog before the change move to the Obsolete lifecycle in the catalog. You can purge obsolete objects from a catalog source. This permanently deletes all objects with the Obsolete lifecycle status, along with associated enrichments, from the catalog.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source from which you want to purge obsolete objects.
4. Click the **Action** menu, and select **Purge Obsolete Objects**.

   A warning message appears on the page.
5. Click **Yes** to confirm.

   A confirmation message appears on the page and the purge job starts. You can monitor the status of the job on the **Monitor** page. If the job succeeds, all obsolete objects present in the catalog source disappear from the catalog.

   Note: If the job fails, some obsolete objects might remain in the catalog. To remove the remaining obsolete objects, run the purge job again.

Curating CLAIRE recommendations

You can view related catalog sources and accept or reject CLAIRE recommendations for a catalog source.

Applicable to catalog sources that generate lineage.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source for which you want to curate CLAIRE recommendations.
4. Hover your mouse over the catalog source and click **View Lineage Discovery** from the **Action** menu.

   Note: The **View Lineage Discovery** option is available for catalog sources that are run with the lineage discovery capability.

   The **Configure** page appears with the catalog source filtered on the **Lineage Discovery** tab of the **Lineage** tab.
   The **Related Catalog Sources** tab displays a list of related catalog sources.
5. Hover your mouse over a related catalog source and click **Accept** or **Reject** from the **Action** menu.

   For more information about lineage discovery and curating CLAIRE recommendations, see *Lineage discovery* in the *Administration* help.

Managing configuration permissions for catalog sources

As a service administrator, you can grant configuration permissions for catalog sources to different roles in your organization, and add specific users and user groups to those roles so that they can perform operational activities on a catalog source.

To manage permissions for catalog source configuration, verify that at least one role has the catalog source configuration **Set Permissions** functionality enabled for Metadata Command Center in Administrator. In Administrator, create the required roles, assign users and user groups to the roles, and specify appropriate permissions to the roles.

You can select roles and add specific users or user groups to those roles to grant the following levels of permission for catalog source configuration:

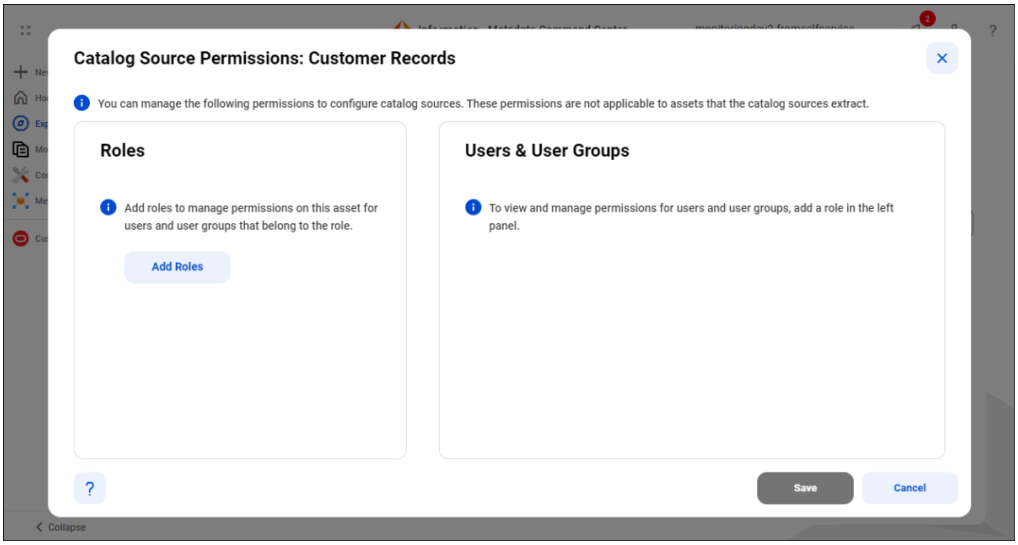
- Create
- Read
- Update
- Run
- Set Permissions

Based on the permissions granted to the roles, the selected users and user groups belonging to that role can perform appropriate operations for configuring catalog sources.

1. Go to the **Explore** page in Metadata Command Center.
2. From the list of catalog sources, search for the catalog source for which you want to manage permissions.

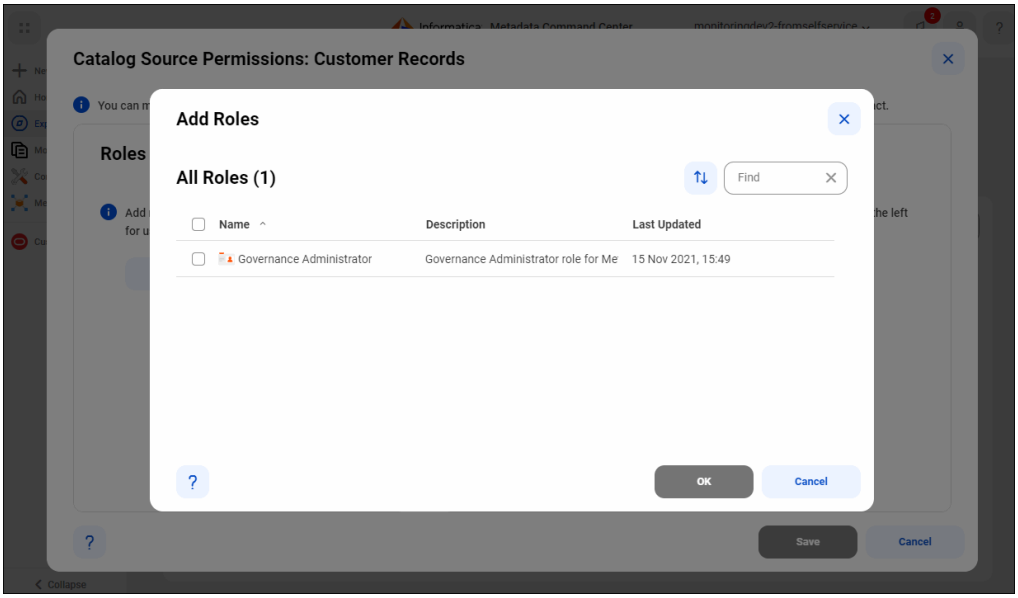
3. Click the **Action** menu, and select **Permissions**.

   The **Catalog Source Permissions** dialog box appears.



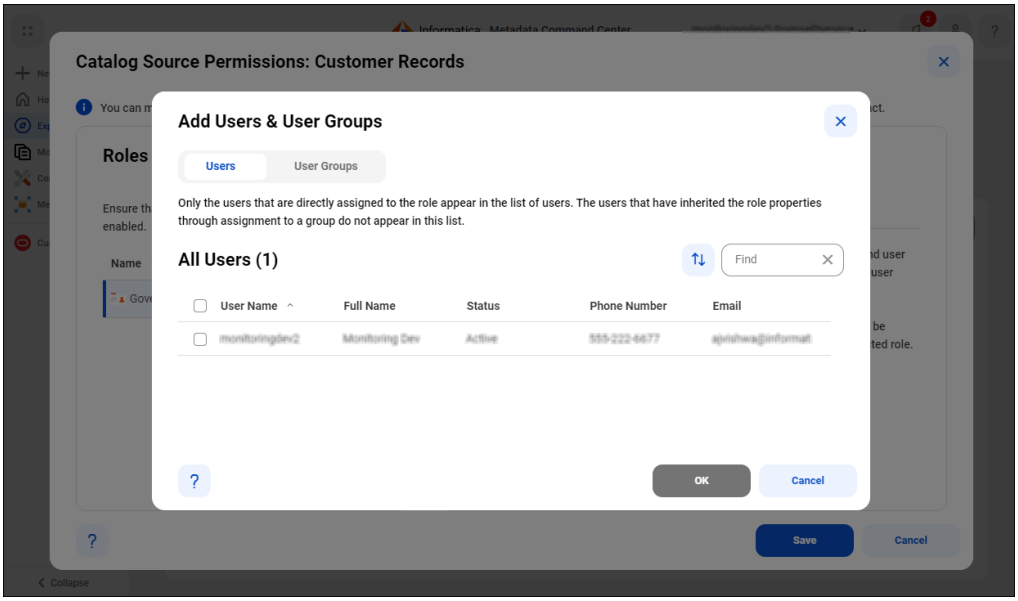
4. In the **Roles** panel, click **Add Roles**.

   The **Add Roles** dialog box appears.



5. Select one or more roles, and click **OK** to add the roles.
6. In the **Add Users & User Groups** panel, by default, all users and user groups that belong to the selected role are granted the role-defined permissions for the catalog source configuration.

To assign permissions only to the selected users and user groups that belong to this role, choose **Specific Users and User Groups**.



7. Select **Add Specific Users and User Groups**.

   The **Add Users & User Groups** dialog box displays the list of users and user groups that the administrator has assigned to the selected role.

8. Select one or more users or user groups to assign to the selected role, and click **OK**.

   Only the selected users and user groups belonging to the specified role will be granted the role-defined permissions for catalog source configuration.

9. View the assigned permissions for each user and or group, and click **Save**.